3.13 Graded Rings

Definition 3.13.1 (Graded Ring over $\mathbb{Z}$)
*A ring A (resp. A-module M) is said to be $\mathbb{Z}$-**graded** (or simply **graded**) if it is graded in the sense of (3.12.2) with gradation group $\Delta = \mathbb{Z}$.*

*If $i < 0 \implies A_i = 0$ (resp. $M_i = 0$) then we say A (resp. M) is **positively graded**.*

Note in this case that we have the multiplication formula

$$(a \cdot b)_i = \sum_{j+k=i} a_j \cdot b_k$$

Definition 3.13.2
*For an element $a \in A$ we define the **degree** to be*

$$\deg(a) \coloneqq \max\{i \mid a_i \neq 0\}$$

Definition 3.13.3 (Essential and Irrelevant Ideals)
*Let A be a positively graded ring. Then we say a homogenous ideal $\mathfrak{a}$ is **irrelevant** if the following condition holds*

$$\exists n_0 \text{ s.t. } n \geq n_0 \implies \mathfrak{a}_n = A_n$$

*In particular A_+ is irrelevant. Otherwise we say that it is **essential**.*

Proposition 3.13.4 (Radical of Homogenous Ideal is Homogenous)
Let A be a graded ring and $\mathfrak{a}$ a homogenous ideal. Then $\sqrt{\mathfrak{a}}$ is also a homogenous ideal.

Proof. Define $\mathfrak{r} \coloneqq \sqrt{\mathfrak{a}}$ and fix $x \in \mathfrak{r}$. Suppose

$$x = \sum_{i=-\infty}^{\infty} x_i$$

where $\delta(x_i) = i$ and $x^k \in \mathfrak{r}$. Then

$$x^k = \sum_{i=-\infty}^{\infty} x_i^k$$

where $\delta\left(x_i^k\right) = ki$. Without loss of generality we may assume $k > 0$ so by assumption $x_i^k \in \mathfrak{r}$ for all i and therefore $x_i \in \sqrt{\mathfrak{r}}$. ∎

Proposition 3.13.5
Let A be a graded ring and $\mathfrak{p}$ a homogenous ideal satisfying the following property

$$a \in A_n, b \in A_m, ab \in \mathfrak{p} \implies a \in \mathfrak{p} \text{ or } b \in \mathfrak{p}$$

Then $\mathfrak{p}$ is a prime ideal.

Proof. Suppose $\mathfrak{p}$ is not prime. Then there exists $a, b \in A$ such that $ab \in \mathfrak{p}$ and $a, b \notin \mathfrak{p}$. Write $a = \sum_j a_j$ and $b = \sum_k b_k$. Then

$$ab = \sum_{i=-\infty}^{\infty} \left(\sum_{(j,k)|j+k=i} a_j b_k \right)$$

As $\mathfrak{p}$ is assumed homogenous then for each i we have

$$(ab)_i \coloneqq \sum_{(j,k)|j+k=i} a_j b_k \in \mathfrak{p}_i$$

Let j' (resp. k') be the largest integer such that $a_{j'} \notin \mathfrak{p}$ (resp. $b_{k'} \notin \mathfrak{p}$). For all other pairs j'', k'' such that $j' + k' = j'' + k''$ we have by maximality $a_{j''} b_{k''} \in \mathfrak{p}_i$. Therefore we conclude $a_{j'} b_{k'} \in \mathfrak{p}$, which contradicts the original assumption. ∎

Proposition 3.13.6
Let A be a positively graded ring. Then the following are equivalent

a) A_+ is a finitely-generated A-module
b) A is a finitely-generated A_0-algebra

More precisely for homogenous elements $x_0, ..., x_n \in A_+$ we have

$$A_+ = \sum_{i=0}^{n} Ax_i \iff A = A_0[x_0, ..., x_n]$$

*In this case we say that A is **finitely generated**. Further*

$$A_+^m = (x_0, ..., x_n)^m$$

for all integers $m > 0$.

Proposition 3.13.7 (Criteria for Irrelevant ideals)
Let A be a positively graded ring which is finitely generated and $\mathfrak{a}$ a homogenous ideal. Then the following are equivalent

a) $\mathfrak{a}$ is irrelevant (i.e. $n \geq n_0 \implies \mathfrak{a}_n = A_n$)
b) $A_+ \subset \sqrt{\mathfrak{a}}$

Proof. a) $\implies$ b) It is sufficient to show that $i > 0 \implies A_i \subset \sqrt{\mathfrak{a}}$. Given $x \in A_i$ then by the Archimedean property there exists $k > 0$ such that $ik > n_0$. Therefore by assumption $x^k \in \mathfrak{a}$ and $x \in \sqrt{\mathfrak{a}}$.

b) $\implies$ a) TODO. ∎

Proposition 3.13.8
Let A be a graded ring and $f \in A_1$. Then there is an isomorphism

$$\begin{aligned} A_{(f)}[X, X^{-1}] &\xrightarrow{\sim} A_f \\ \sum_{i\in\mathbb{Z}} \frac{a_{n_i}}{f^{n_i}} X^i &\longrightarrow \sum_{i\in\mathbb{Z}} \frac{a_{n_i}}{f^{n_i - i}} \quad a_{n_i} \in A \text{ homogenous of degree } n_i \\ \sum_{i\in\mathbb{Z}} \frac{a_i}{f^i} X^{i-r} &\longleftarrow \frac{\sum_{i\in\mathbb{Z}} a_i}{f^r} \end{aligned}$$

Proof. Denote the maps by ψ, ϕ, then ψ exists and is a ring homomorphism by (3.9.2) satisfying $\psi(X) = f$ and whence $\psi(X^i) = f^i$. To show ψ is well-defined first consider the map

$$\begin{aligned} \psi' : A &\to A_{(f)}[X, X^{-1}] \\ \sum_{i\in\mathbb{Z}} a_i &\to \sum_{i\in\mathbb{Z}} \frac{a_i}{f^i} X^i \end{aligned}$$

which is evidently a well-defined homomorphism of abelian groups. Furthermore

$$\left(\sum_{i\in\mathbb{Z}} \frac{a_i}{f^i} X^i\right)\left(\sum_{i\in\mathbb{Z}} \frac{b_i}{f^i} X^i\right) = \sum_{i\in\mathbb{Z}} \left(\sum_{j+k=i} \frac{a_j b_k}{f^{j+k}}\right) X^i = \sum_{i\in\mathbb{Z}} \left(\frac{1}{f^i} \sum_{j+k=i} a_j b_k\right) X^i$$

and so the map is a ring-homomorphism by the multiplication formula in A. By the universal property of localisation the ring homomorphism ϕ exists with $\phi(f) = X$ and hence $\phi(f^i) = X^i$. We have

$$\phi(\psi(X)) = \phi(f) = X$$

$$\phi\left(\psi\left(\frac{a_j}{f^j}\right)\right) = \phi\left(\frac{a_j}{f^j}\right) = \frac{a_j}{f^j}$$

so by linearity $\phi \circ \psi = \mathbf{1}$. Similarly

$$\psi\left(\phi\left(\frac{a_i}{f^r}\right)\right) = \psi\left(\frac{a_i}{f^i} X^{i-r}\right) = \frac{a_i}{f^i} f^{i-r} = \frac{a_i}{f^r}$$

so by linearity $\psi \circ \phi = \mathbf{1}$. ∎